

Network Working Group
Internet-Draft
Obsoletes: 7451 (if approved)
Intended status: Best Current Practice
Expires: 16 October 2026

S. Hollenbeck
Verisign Labs
14 April 2026

Commenté [MB1]: I think Standard Track would be more appropriate.

Extension Registry for the Extensible Provisioning Protocol (EPP) draft-ietf-regext-ext-registry-epp-06

Abstract

The Extensible Provisioning Protocol (EPP) includes features to add new functionality by extending the protocol. It does not, however, describe how those extensions are ~~maintained~~managed. This document describes a procedure for the registration and ~~management~~maintenance of EPP extensions ~~to EPP~~, and ~~it~~ specifies a format for an IANA registry to record those extensions. ~~If approved, t~~This document obsoletes RFC 7451.

Commenté [MB2]: To avoid this being confused with management in OPS :-)

Status of This Memo

This Internet-Draft is submitted in full conformance with the provisions of BCP 78 and BCP 79.

Internet-Drafts are working documents of the Internet Engineering Task Force (IETF). Note that other groups may also distribute working documents as Internet-Drafts. The list of current Internet-Drafts is at <https://datatracker.ietf.org/drafts/current/>.

Internet-Drafts are draft documents valid for a maximum of six months and may be updated, replaced, or obsoleted by other documents at any time. It is inappropriate to use Internet-Drafts as reference material or to cite them other than as "work in progress."

This Internet-Draft will expire on 16 October 2026.

Copyright Notice

Copyright (c) 2026 IETF Trust and the persons identified as the document authors. All rights reserved.

This document is subject to BCP 78 and the IETF Trust's Legal Provisions Relating to IETF Documents (<https://trustee.ietf.org/license-info>) in effect on the date of publication of this document. Please review these documents carefully, as they describe your rights and restrictions with respect to this document. Code Components extracted from this document must include Revised BSD License text as described in Section 4.e of the Trust Legal Provisions and are provided without warranty as described in the Revised BSD License.

Table of Contents

1. Introduction 2
1.1. Conventions Used in This Document 3
2. Extension Specification and Registration Procedure 3
2.1. Extension Specification 3
2.1.1. Designated Expert Evaluation Criteria 4
2.2. Registration Procedure 5
2.2.1. Required Information 5
2.2.2. Registration Form 6
2.2.3. Registration Processing 8
2.2.4. Updating Registry Entries 8
3. IANA Considerations 8
4. Security Considerations 9
5. References 9
5.1. Normative References 9
Acknowledgements 11
Change Log 11
Author's Address 12

1. Introduction

Domain name registries implement a variety of operational and business models. The differences in these models make it impossible to develop a "one size fits all" provisioning protocol; the Extensible Provisioning Protocol (EPP) [STD69] was designed to focus on a minimal set of common functionality with built-in extension capabilities that allow new features to be specified on an "as needed" basis. Guidelines for extending EPP are documented in RFC 3735 [RFC3735].

[RFCs-3735] and [RFC5730] do not describe how extension development can be managed and coordinated. This has led to a situation in which server operators can develop different extensions to address similar needs, such as the provisioning of Value Added Tax (VAT) information. Clients then need to support multiple extensions that serve similar purposes, and interoperability suffers as a result.

[RFC7451] filled that gap by defining An-the Extensions for the Extensible Provisioning Protocol (EPP) IANA registry [REF] can be used to help manage and coordinate the development of protocol-EPP extensions. This document describes an IANA registry that will be used to coordinate the development of EPP extensions.

This update was-is written to address a few issues that were identified with RFC-7451 [RFC7451] over time. Refer to Section 1.1 for more details.

1.1. Changes Since RFC 7451

The main changes since [RFC7421] are as follows:

* The intended status was changed to xxxx.

Commenté [MB3]: Cite as references

Commenté [MB4]: Please add a new reference entry that points to https://www.iana.org/assignments/epp-extensions/epp-extensions.xhtml

Commenté [MB5]: As the registry exists already.

Commenté [MB6]: Also helps reference the main changes

Commenté [MB7]: The changes about XML validation, requiring IETF prefix, etc. are not reflected here.

Commenté [MB8]: To be completed

* The name of the mailing list used to review and discuss registration requests was changed from "eppext" to "regext"~~throughout the document~~.

* Text has been added to describe ~~Reviewer~~Designated Expert responsibility to confirm correctness of URIs used in extension registration requests.

* "Other" has been added to the set of document status values for the registry to avoid confusion with "Informational" RFCs.

* Section 2.2.3 has been updated to note that registry entries can be removed with IESG approval.

* Section 2.2.2 has been updated by changing "<registrant name>, <email address>" to "<name>, <address>" to meet right margin constraints.

Commenté [MB9]: Reorder those to follow the sections order

1.1. Conventions Used in This Document

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 [RFC2119] [RFC8174] when, and only when, they appear in all capitals, as shown here.

2. Extension Specification and Registration Procedure

This section describes the format of ~~an the Extensions for the Extensible Provisioning Protocol (EPP)~~ IANA registry ~~[REF]~~ and the procedures used to populate and manage registry entries.

Commenté [MB10]: Link to the IANA registry

2.1. Extension Specification

This registry uses the "Specification Required" policy described in ~~RFC-8126~~Section 4.6 of [RFC8126]. An English language version of the extension specification ~~will~~ be referenced from the registry, though non-English versions of the specification may also be provided. Note that Section 2.1 of ~~RFC-3735~~ [RFC3735] provides specific guidelines for documenting EPP extensions.

Commenté [MB11]: Should? Must?

The "Specification Required" policy requires review by a designated expert. Section 5 of ~~RFC-8126~~ [RFC8126] describes the role of designated experts and the function they perform. This policy also requires "a permanent and readily available public specification". RFC documents meet that requirement. Proprietary specifications may meet that requirement, depending on how they are archived and accessible. ~~Internet-Draft documents do not meet that requirement.~~ ~~RFC-2026~~ [RFC2026] notes that "Internet-Drafts have no formal status, and are subject to change or removal at any time".

Commenté [MB12]: This repeating text already known from 8126

Commenté [MB13]: Some registries accepts I-Ds. I think that we need simply say that we don't accept them for EPP without making a general claim.

Commenté [MB14]: I would delete.

2.1.1. Designated Expert Evaluation Criteria

A high-level description of the role of the designated expert is described in Section 5.2 of ~~RFC-8126~~ [RFC8126]. Specific guidelines

for the appointment of designated experts and the evaluation of EPP extensions are provided here.

The IESG should appoint a small pool of individuals (perhaps 3 - 5) to serve as designated experts, as described in Section 5.2 of RFC 8126 [RFC8126]. The pool should have a single administrative chair who is appointed by the IESG. The designated experts MUST use the existing regext mailing list (regext@ietf.org) or its successor for public discussion of registration requests.

Extensions should be evaluated for architectural soundness using the guidelines described in RFC 3735 [RFC3735] (or its future replacement), including the Security Considerations section of that document. Expert evaluation should explicitly include consideration of the privacy consequences of proposed extensions, and, at a minimum, ensure that any privacy considerations are fully documented in the relevant specification(s).

URIs proposed in extensions (XML namespace and schema registration requests are commonly found in EPP extensions) should be evaluated for both syntactic and semantic correctness. XML schemas, XML schema URIs, and XML namespace URIs defined in the extension specification MUST be registered in the IETF XML Registry using the procedures described in RFC 3688 [RFC3688]. IETF namespaces MUST be reserved for IETF specifications published in the IETF Stream. Non-IETF namespaces MUST be used for non-IETF specifications (i.e., specification published in other streams such as which includes RFC documents published using the Independent Submission stream); the designated experts may need to work with a registrant to identify URIs that can be added to the IETF XML Registry. Extensions and any normative reference necessary to implement the extension MUST NOT be denoted with "work in-progress" or any similar description.

The results of the evaluation MUST be shared via email with the registrant and the regext mailing list (or its successor). Issues discovered during the evaluation can be corrected by the registrant, and those corrections can be submitted to the designated experts until the designated experts explicitly decide to accept or reject the registration request. The designated experts MUST make an explicit decision and that decision MUST be shared via email with the registrant and the regext mailing list (or its successor). The decision MUST include justifications that motivated the decision.

If the specification for an extension is an IETF Standards Track document, no review is required by the designated expert.

Designated experts should be permissive in their evaluation of requests to register extensions that have been implemented and deployed by at least one registry/registrar pair. This implies that it may indeed be possible to register multiple extensions that provide the same functionality. Requests to register extensions that have not been deployed should be evaluated with a goal of reducing functional duplication. A potential registrant who submits a request to register a new, un-deployed extension that includes similar functionality to an existing, registered extension should be made

Commenté [MB15]: I see the registry having primary/secondary indication. Is that something the we need to call out here?

As I don't see any specific provision here, I expect all DEs to be on the same level.

Commenté [MB16]: What are the responsibilities of chairs vs other DEs?

Commenté [MB17]: Won't age well.

Commenté [MB18]: Split the paragraph for better readability and identification of criteria

Commenté [MB19]: Add clarity

Commenté [MB20]: For consistency with the DE discussion above.

Commenté [MB21]: Position this as a sperate item.

aware of the existing extension. The registrant should be asked to reconsider their request given the existence of a similar extension. Should they decline to do so, perceived similarity SHOULD NOT be a sufficient reason for rejection as long as all other requirements are met.

In cases where a registration decision could be perceived as creating a conflict of interest for a particular Designated Expert, that Expert SHOULD defer to the judgment of the other Experts.

2.2. Registration Procedure

The registry contains information describing each registered extension. Registry entries are created and managed by sending forms to IANA that describe the extension and the operation to be performed on the registry entry.

2.2.1. Required Information

Name of Extension: A case-insensitive, ASCII text string that contains the name of the extension specification. Non-ASCII representations of the extension name can be included in the "Notes" described below.

Document Status: The document status of the specification document. For RFC documents, the possible set of values includes "Standards Track", "Informational", "Experimental", "Historic", and "BCP" as described in Sections 4 and 5 of ~~RFC-2026~~ [RFC2026]. For documents that are not RFCs, this will always be "Other".

Reference: A permanent, publicly available reference to the specification of this extension. This could be an RFC number or some other pointer to the document defining the extension that meets the "Specification Required" registry policy.

Registrant Name and Email Address: The name and email address of the person that is responsible for managing the registry entry. If the extension is registered by an IETF stream RFC, this can simply be listed as "IETF, <iesg@ietf.org>".

TLDs: A text string containing the top-level domain name (or domain names), including the preceding ".", for which the extension has been specified (e.g., ".org"). If there are multiple TLDs, they are given as a list of domain names separated by commas, (e.g. ".com", ".net"). Internationalized Domain Name (IDN) TLDs MUST be specified in A-label [RFC5890] format. If the extension is not associated with a specific top-level domain, the case-insensitive text string "Any" can be used to indicate that. If the extension is not associated with domain name processing, the case-insensitive text string "N/A" (Not Applicable) can be used to indicate that.

IPR Disclosure: A pointer to any Intellectual Property Rights (IPR) disclosure document(s) related to this extension, or "None" ~~MAY-may~~ be used if there are no such disclosures. This can be an IPR disclosure filed with the IETF in accordance with ~~RFC-8179~~ [RFC8179] [BCP79] if the extension is part of an IETF Contribution, or it can be other IPR

Commenté [MB22]: Should we say that the name should not overlap with already registered extensions?

Commenté [MB23]: -- I see this not followed in the registry. For example:

Balance Mapping for the Extensible Provisioning Protocol (EPP) Informational [http://www.verisign.com/assets/epp-sdk/verisign_epp-extension_balance_v00.html] [VeriSign_Inc.] AnyNoneActiveNone

-- We need to add an IANA action to update all such entries to follow the guidance.

Commenté [MB24]: I think that we need to point to 2.1.1 where we set the expectation for EPP context.

Commenté [MB25]: -- Should this be changed to «entity» as I see in the registry cases where the contact is not a person.

Commenté [MB26]: I don't think the normative language is needed here.

Commenté [MB27]: Better to use BCP label to inherit any future revisions of 8179

disclosure documents identifying the claimed intellectual property rights and terms of use for extensions that are not part of an IETF Contribution.

Status: Either "Active" or "Inactive". The "Active" status is used for extensions that are currently implemented and in use. The "Inactive" status is used for extensions that are not implemented or are otherwise not being used. "Inactive" can also be used for extensions for which a reference specification becomes unavailable as described in Section 2.2.4.

Notes: Either "None" or other text that describes optional notes to be included with the registered extension. If the Status value is "Inactive", text MUST be included to describe how and when this state was reached.

Commenté [MB28]: How this is determined? How this is maintained?

Is this purely declarative?

Commenté [MB29]: -- I think this applies for any state change. The other transition is also interesting to track.

Commenté [MB30]: Do we expect that formatting issues will lead to reject a request?

2.2.2. Registration Form

The required information MUST be formatted consistently using the following registration form template in Figure 1. Form field names and values MAY appear on the same line.

```
-----BEGIN FORM TEMPLATE-----
Name of Extension: <text string> (quotes are optional)

Document Status: <document status>

Reference: <RFC number, URL, etc.>

Registrant Name and Email Address: <name>, <address>

TLDs: "Any"|"N/A"|<one or more TLD text strings separated by commas>

IPR Disclosure: "None"|"URL"

Status: "Active"|"Inactive"

Notes: "None"|"optional text"

-----END FORM TEMPLATE-----
```

Figure 1: Registration Form Template

An ~~Example-example~~ form with RFC specification is provided in Figure 2+.

```
-----BEGIN FORM-----
Name of Extension:
"An Extension RFC for the Extensible Provisioning Protocol (EPP)"

Document Status: Standards Track

Reference: RFC XXXX

Registrant Name and Email Address: IETF, <iesg@ietf.org>

TLDs: Any
```

IPR Disclosure: None

Status: Active

Notes: None

-----END FORM-----

Commenté [MB31]: I know this is an example, but do we require the extension to be implemented/deployed?

Figure 2: Example Registration Form in an RFC

An ~~Example-example~~ form with non-RFC specification is depicted in Figure 3+. This example includes an IPR disclosure.

-----BEGIN FORM-----

Name of Extension:

"An Example Extension for the .example Top-Level Domain"

Document Status: Other

Reference:

<https://www.example.com/html/example-epp-ext.txt>

Registrant Name and Email Address: John Doe, jdoe@example.com

TLDs: .example

IPR Disclosure:

<https://www.example.com/ipr/example-epp-ext-ipr.html>

Status: Active

Notes: None

-----END FORM-----

Figure 3: Example Registration Form with a Non-RFC Specification.

2.2.3. Registration Processing

Registrants should send each registration form to IANA with a single record for incorporation into the registry. Send the form via email to [<iana@iana.org>](mailto:iana@iana.org) or complete the online form found on the IANA web site. The subject line MUST indicate whether the enclosed form represents an insertion of a new record (indicated by the word "INSERT" in the subject line), replacement of an existing record (indicated by the word "MODIFY" in the subject line), deactivation of an existing record (indicated by the word "DEACTIVATE" in the subject line), or removal of an existing record (indicated by the word "REMOVE" in the subject line).

Commenté [MB32]: Subject missing

~~IESG Approval (Section 4.10 of [RFC8126]) is REQUIRED to remove or deactivate Registrations created through IETF consensus can only be removed or deactivated with IESG Approval (see [RFC8126]).~~

Commenté [MB33]: Use normative language for better clarity

Registrations not created through IETF consensus can be

removed or deactivated ~~with-but~~ the approval of the IESG is REQUIRED,
in consultation
with or at the request of the Designated Experts. Registrations not
created through IETF consensus can also be removed or deactivated by
the original registrant, in consultation with the Designated Experts.

On receipt of a registration request, IANA will initiate review by
the designated expert(s), who will evaluate the request using the
criteria in Section 2.1.1 in consultation with the current working
group mailing list focused on the development of EPP extensions.

Commenté [MB34]: -- How to make sure this change
does not impact uses by other entities than the
registrant?

There might be operational considerations out there. See
below

Commenté [MB35]: There may not exist such WG

2.2.4. Updating Registry Entries

When submitting changes to existing registry entries, include text in
the "Notes" field of the registration form describing the change.
Under normal circumstances, registry entries are only to be updated
by the registrant. If the registrant becomes unavailable or
otherwise unresponsive, the-a designated expert can submit a
registration form to IANA to update the registrant information.

Commenté [MB36]: Subject missing

Commenté [MB37]: As there are many

Entries can change state from "Active" to "Inactive" and back again
as long as state-change requests conform to the processing
requirements identified in this document. In addition to entries
that become "Inactive" due to a lack of implementation, entries for
which a specification becomes consistently unavailable over time
should be marked "Inactive" by the designated expert until the
specification again becomes reliably available.

3. IANA Considerations

IANA has created the "Extensions for the Extensible Provisioning
Protocol (EPP)" registry to manage EPP extensions. This registry has
its own heading on IANA's protocol listings. The information to be
registered and the procedures to be followed in populating the
registry are described in Section 2.

Name of registry: Extensions for the Extensible Provisioning Protocol
(EPP)

Section at <https://www.iana.org/protocols>:

Registry Title:

Extensions for the Extensible Provisioning Protocol (EPP)

Registry Name:

Extensions for the Extensible Provisioning Protocol (EPP)

Registration Procedure:

Specification Required

Reference:

This document

Required information:

See Section 2.2.1

Review process:

"Specification Required" as described in RFC 8126 [RFC8126]

Size, format, and syntax of registry entries:

See Section 2.2.1

Initial assignments and reservations:

Preserved from the existing registry. Please change all non-
RFC entries in the registry that have document status
"Informational" to document status "Other".

Commenté [MB38]: This is weird as we are no creating a
new registry.

In addition, the form used to populate and manage the registry has been added to the table of Protocol Registration Forms maintained by IANA. IANA is further requested to forward all designated expert review requests to both the designated expert and the "regext" mailing list or its successor.

Commenté [MB39]: I would delete this part of the text as the registry exists.

This document requests IANA to replace the reference of the EPP registry to the RFC number to be assigned to this document.

X. Operational Considerations

The updates defined in this document are meant to enhance the guidance for reviewing EPP extensions (e.g., avoid squatting IETF namespace by non-IETF extensions) and thus allows for better interoperability.

Section 3 updates the content of "Document Status" to be compliant with the guidance in Section 2. That update does not impact the extension itself. No operational issues are induced by that update.

Section 2 includes provisions for modifying and deleting existing registration entries by a registrant. Such requests MUST NOT be granted if the requested action has operational implications on other entities that deploy that extension.

4. Security Considerations

This document introduces no new security considerations to EPP. However, extensions should be evaluated according to the Security Considerations of ~~RFC-3735~~ [RFC3735] and ~~STD-69~~ [STD69].

5. References

5.1. Normative References

- [STD69] Internet Standard 69,
<<https://www.rfc-editor.org/info/std69>>.
At the time of writing, this STD comprises the following:
- Hollenbeck, S., "Extensible Provisioning Protocol (EPP)",
STD 69, RFC 5730, DOI 10.17487/RFC5730, August 2009,
<<https://www.rfc-editor.org/info/rfc5730>>.
- Hollenbeck, S., "Extensible Provisioning Protocol (EPP)
Domain Name Mapping", STD 69, RFC 5731,
DOI 10.17487/RFC5731, August 2009,
<<https://www.rfc-editor.org/info/rfc5731>>.
- Hollenbeck, S., "Extensible Provisioning Protocol (EPP)
Host Mapping", STD 69, RFC 5732, DOI 10.17487/RFC5732,
August 2009, <<https://www.rfc-editor.org/info/rfc5732>>.
- Hollenbeck, S., "Extensible Provisioning Protocol (EPP)
Contact Mapping", STD 69, RFC 5733, DOI 10.17487/RFC5733,
August 2009, <<https://www.rfc-editor.org/info/rfc5733>>.
- Hollenbeck, S., "Extensible Provisioning Protocol (EPP)
Transport over TCP", STD 69, RFC 5734,

- DOI 10.17487/RFC5734, August 2009,
<<https://www.rfc-editor.org/info/rfc5734>>.
- [RFC2026] Bradner, S., "The Internet Standards Process -- Revision 3", BCP 9, RFC 2026, DOI 10.17487/RFC2026, October 1996,
<<https://www.rfc-editor.org/info/rfc2026>>.
- [RFC2119] Bradner, S., "Key words for use in RFCs to Indicate Requirement Levels", BCP 14, RFC 2119, DOI 10.17487/RFC2119, March 1997,
<<https://www.rfc-editor.org/info/rfc2119>>.
- [RFC3688] Mealling, M., "The IETF XML Registry", BCP 81, RFC 3688, DOI 10.17487/RFC3688, January 2004,
<<https://www.rfc-editor.org/info/rfc3688>>.
- [RFC3735] Hollenbeck, S., "Guidelines for Extending the Extensible Provisioning Protocol (EPP)", RFC 3735, DOI 10.17487/RFC3735, March 2004,
<<https://www.rfc-editor.org/info/rfc3735>>.
- [RFC5890] Klensin, J., "Internationalized Domain Names for Applications (IDNA): Definitions and Document Framework", RFC 5890, DOI 10.17487/RFC5890, August 2010,
<<https://www.rfc-editor.org/info/rfc5890>>.
- [RFC7451] Hollenbeck, S., "Extension Registry for the Extensible Provisioning Protocol", RFC 7451, DOI 10.17487/RFC7451, February 2015, <<https://www.rfc-editor.org/info/rfc7451>>.
- [RFC8126] Cotton, M., Leiba, B., and T. Narten, "Guidelines for Writing an IANA Considerations Section in RFCs", BCP 26, RFC 8126, DOI 10.17487/RFC8126, June 2017,
<<https://www.rfc-editor.org/info/rfc8126>>.
- [RFC8174] Leiba, B., "Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words", BCP 14, RFC 8174, DOI 10.17487/RFC8174, May 2017, <<https://www.rfc-editor.org/info/rfc8174>>.
- [RFC8179] Bradner, S. and J. Contreras, "Intellectual Property Rights in IETF Technology", BCP 79, RFC 8179, DOI 10.17487/RFC8179, May 2017,
<<https://www.rfc-editor.org/info/rfc8179>>.

Commenté [MB40]: Please move this one to Informative as it will be obsoleted.

Acknowledgements

The information described in the registry is based on a suggestion posted to the provreg mailing list by Jay Daley in August 2013. The need to update RFC 7451 was first proposed by Gavin Brown. Additional feedback for the update was provided by the following people: Gavin Brown, James Galvin, James Gould, Pawel Kowalik, Andrew Newton, and Jasdip Singh.

Change Log

This section is to be removed before publishing as an RFC.

- 00: Initial WG version.
- 01: WG last call edits: added reference to RFC 2026 to clarify the status of Internet-Draft documents as extension specifications. "IESG approval" -> "IESG Approval" in Section 2.2.3. Added DEACTIVATE and REMOVE request processing to Section 2.2.3. Clarified use of IETF namespaces and "work in progress" specifications in Section 2.2.1. Clarified status values in Section 2.2.1. Updated acknowledgements.
- 02: Changed intended status from Informational to BCP. Added text to address Independent Submission stream RFCs to Section 2.1. Noted that the value of the TLDs field (Section 2.2.1) can be "N/A". Added text to Section 2.2.1 to ensure that it's consistent with Section 2.2.4. Updated examples to use "https" instead of "http". Updated acknowledgements.
- 03: Updated to use BCP 14 keywords. Updated Section 2.1 and Section 2.1.1 to clarify ISE RFC use as a reference specification for an extension.
- 04: Second WG last call edits: Updated "XML schemas, XML schema URIs, and XML namespace URIs" wording in Section 2.2.1. Noted that the value of "TLDs" can be "N/A" in Section 2.2.2. Updated "removed or deactivated" wording in Section 2.2.3. Changed RFC 3735 from an informative reference to a normative reference. Changed "IESG" to "IETF" in the "Registrant Name and Email Address" description in Section 2.2.1 and the example registration template in Section 2.2.2. Updated obsolete normative references to RFCs 3979 and 4879 (obsoleted by RFC 8179).
- 05: Post-WG last call edits: changed "should not" to "SHOULD NOT" in the last sentence of Section 2.1.1. Changed "RFC 5730" to "STD 69" in Section 1 and added "STD 69" to Section 4.
- 06: Shepherd write-up edit: Added "If approved" sentence to the Abstract.

Author's Address

Scott Hollenbeck
Verisign Labs
12061 Bluemont Way
Reston, VA 20190
United States of America
Email: sah@sahollenbeck.com, shollenbeck@verisign.com
URI: <https://www.verisignlabs.com/>